

Aditya Tayal

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Final-year CSE undergraduate at IIT Mandi who ships software people actually use. Solo-built PlanMyDegree, the degree planner **1,741 IIT Mandi students** use, plus exam, leave and lab systems for the institute. Five startup internships, **All India Rank 1** in an IEEE ICASSP challenge, 20+ hackathons, and elected **UG Academic Affairs Secretary** for ~3,500 undergraduates.

EDUCATION

Indian Institute of Technology Mandi

B.Tech in Computer Science & Engineering | CGPA: 8.80

Mandi, Himachal Pradesh

Aug 2023 – May 2027 (expected)

Coursework: Machine Learning, Deep Learning, Generative AI, Data Science, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Information Security, Optimization

Ramakrishna Mission

Class XII, PCM with Computer Science | 89.40%

Gwalior, Madhya Pradesh

2020 – 2022

Wendy School Junior College

Class X, ICSE | 98.20%

Gwalior, Madhya Pradesh

2007 – 2020

EXPERIENCE

Major Technical Project, American Express

Imbalanced fraud detection | Python, scikit-learn, XGBoost, LightGBM, CatBoost

Aug 2026 – Present

Team project

- Stress-testing Self-Paced Ensembles on a multi-product fraud portfolio built from public data.
- Compared Decision Tree, XGBoost, LightGBM and CatBoost learners, pooled vs per-product, over 3 seeds.
- Built a cost model that turns a fraud ranking into money saved, alongside AUPRC, F1 and Recall@K.
- Best setup recovers **64%** of recoverable fraud loss; results written up as a full report.

Software Developer Intern, MagoStays

Koha Consultants Pvt. Ltd. | TypeScript, Node.js, PostgreSQL, Next.js, React Native

Sep 2026 – Present

Travel tech

- Full-stack developer on a villa and farmstay booking platform: backend, admin, website and host app.
- Own payments, invoicing and booking features end to end, from database schema to user interface.
- Write unit, integration and end-to-end tests, plus a design doc for every flow I own.
- Work with product and finance on edge cases, audit trails and safe database migrations.

Software Developer Intern, InstaFarms

VWA Hospitality Pvt. Ltd. | TypeScript, Node.js, PostgreSQL, Next.js

Apr 2026 – Aug 2026

Travel tech

- Full-stack developer on a farmstay and villa booking platform across the backend, admin and website.
- Built booking, pricing and tax features end to end in a TypeScript monorepo with documented APIs.
- Integrated partner inventory into checkout and sped up the availability calendar with batched calls.
- Added a new property type across all apps and improved the website's search and SEO pages.

AI/ML Engineer Intern, WattCycle

Solar energy | Python, FastAPI, scikit-learn, XGBoost, Prophet, PostgreSQL, React

Dec 2025 – Jun 2026

- Worked across ML and full stack on WattCycle Atlas, a solar-plant monitoring platform.
- Built fault detection that pairs rule-based health checks with Isolation Forest anomaly scoring.
- Designed multi-horizon energy forecasting with XGBoost and Prophet, served through FastAPI.
- Built the analytics backend behind its dashboards; containerised services with Docker and CI.

Full Stack Developer Intern, Garudakshak Technologies

Defence tech | Tauri, Go, React, TypeScript, PostgreSQL

Oct 2025 – Apr 2026

- Built the operator desktop app for a counter-drone system from an empty repository to MVP.
- Cross-platform Tauri app with a Go backend and a React + TypeScript interface.
- Real-time dashboards over WebSockets, role-based access control and automated release builds.
- Worked with hardware and ML teammates to bring their detection models into the operator UI.

Lead Developer Intern, Strive Partners

Hiring tech | React, TypeScript, JavaScript, Django REST, AWS

Sep 2025 – Jun 2026

- Led development of Strive Hunt, a hiring platform for candidates, employers and recruiters.
- Built the candidate and employer portals and an internal applicant-tracking system.
- Real-time shared candidate sheets, bulk Excel import and candidate search with rich filters.
- Secured it with token refresh, CSRF protection and input sanitising; shipped via Docker and CI.

PROJECTS

PlanMyDegree, Academic Planner (Solo)

Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, PWA

Jan 2026 – Present

- Solo-built the degree planner **1,741 IIT Mandi students** use to plan credits, minors and timetables.
- **150+ sign-ups in 36 hours** from one WhatsApp message; 17,555 course enrollments tracked since.
- Pre-registration with seat caps, timetable clash detection, transcript OCR and calendar export.
- 30 data models, ~65 API routes and 880+ commits, rated 4.9/5 in student feedback.

Camouflaged Object Detection (Paid Research)

PyTorch, mmdetection, DINOv2, LoRA, SAM2

Apr 2026 – Jul 2026

- Adapted PAIS (ICCV 2023) to camouflaged objects with only 10% labels: **+7.8 AP** over supervised.
- Designed a text-guided DINOv2 + LoRA decoder: S-measure **0.847** on COD10K, 0.865 on NC4K.
- Reproduced UCOD-DPL (CVPR 2025) with SAM2 refinement: **0.8407 vs the paper's 0.835**.
- Beat the reproduced GBNet baseline on **42/42** dataset-metric checks over 71 experiments.

Domain Unlearning in CLIP (Team of 4)

PyTorch, CLIP ViT-B/16, prompt learning

[Code](#)

Jan 2026 – May 2026

- Made CLIP forget a whole visual domain and beat the published result (**72.10 vs 69.96** H-score).
- Found the forget signal is outvoted N:1 by the kept domains, and rebalanced the loss to fix it.
- Adapted SCRUB from samples to whole domains: **83.6** H-score, **+13.6** points over the paper.

ALS Severity from Voice, SAND 2025 (Team of 4)

Python, audio ML

Oct 2025 – Nov 2025

- Graded ALS severity from speech for an IEEE ICASSP challenge: **All India Rank 1, Global Rank 11**.
- Macro-F1 0.5430 on 5-class severity; India #2 on the disease-progression task.

Exodia 2026 Ticketing & Gate System (Solo)

Next.js, tRPC, Prisma, PostgreSQL, Nodemailer

[Code](#)

Mar 2026 – Apr 2026

- QR ticketing, mailing and gate scanning for the fest's biggest night: **300+ tickets, 200 entries**.
- Single-use database claims stopped **26 reuse attempts**, with **0 duplicate entries** at the gate.
- Throttled multi-account mailer with open tracking; **13 commits** shipped live on event day.

Faculty Leave Portal (Solo)

Fastify, Prisma, React, SQLite, Docker

Sep 2026 – Present

- Leave management for IIT Mandi faculty, built from the institute's own requirements document.
- No stored balances: every figure comes from an append-only ledger the database refuses to edit.
- **324 requirements** traced to **321 automated tests**; OTP login, CSRF guards and upload checks.

TestPlatform, Online Examinations (Solo)

Next.js, Express, Socket.IO, PostgreSQL, Redis

Aug 2026 – Present

- An exam system where the clock lives on the server: Redis-held timers, live rooms, pause and resume.
- Shuffled papers per student, auto-grading for MCQ and numeric answers, passwordless OTP login.
- Hardened after a self-audit: row-locked submits, IDOR fixes and patched CVEs.

Vehicle Counting with Deep Learning Banned (Solo)

Python, OpenCV, NumPy

Jan 2026

- Counted traffic with classical CV only: a 9-stage pipeline built in 48 hours, **rank 10 against teams**.
- Background subtraction, centroid tracking and track merging; **MAE 16.4** on the hidden test set.

MahaDrishti AI, Satellite Crop Verification (Team of 3)

Sentinel-1/2, Prithvi-EO, Random Forest, FastAPI

May 2026

- Checks a farmer's declared crop against satellite imagery plot by plot, with reports in Marathi.
- **Finalist**, Pune International Agri Hackathon; 194 plots per village in under 2 minutes.

Product Price Prediction, Amazon ML Challenge (Team of 3)

Python, LightGBM, TF-IDF

Oct 2025

- Priced 75,000 products from text alone with a 21-model LightGBM ensemble (7 folds, 3 seeds).
- Cut SMAPE from **54.38% to 32.33%** out-of-fold, among 82,787 registered teams.

FinFabric, IDBI Innovate 2026 (Team Lead)

JavaScript, scoring models, Vercel

[Code](#)

Jul 2026

- Led 5 live banking prototypes: MSME health scores, SIP planning, lead scoring and default alerts.

Sahayak, KSP Datathon 2026 (Solo)

Qwen 2.5 Coder, FastAPI, React

[Code](#)

Jun 2026

- Ask a crime database questions in English or Kannada; get back the SQL, source rows and legal codes.
- A read-only SQL guard means a generated query can never change data.

Deepfake Detection (Team of 6)

CLIP, DINOv2, SVM (cuML)

Feb 2025 – May 2025

- Caught images from **19 generators** with frozen vision-transformer features and a GPU SVM.

– 89.4% accuracy across 80,353 test images.	
CaptureSmart, AI Camera Assistant (Team of 6)	Code
<i>Kotlin, CameraX, TensorFlow, Flask</i>	<i>May 2025</i>
– Android camera that scores blur live (MobileNetV2, Pearson 0.89 on 24,000 images).	
– Suggests ISO and shutter speed from 7 image features; 3rd of 60+ teams .	
Adobe India Hackathon 2025 (Team Leader)	1A 1B
<i>Python, PyMuPDF, MiniLM, Docker</i>	<i>Jul 2025</i>
– Top 1.5% of 262,120 teams: multilingual PDF outlines in an offline, CPU-only container.	
– Persona-aware section ranking with MiniLM embeddings, 7 PDFs in under 45 seconds.	
DeepMatch, India Runs Hackathon (Solo)	Code
<i>Python, sentence-transformers</i>	<i>Jun 2026</i>
– Ranks 100K candidates to a shortlist in ~75 s on a CPU ; six heuristics flag fake profiles.	
SevaVoice, HACKHAZARDS '26 (Solo)	Live Code
<i>React, TypeScript, Express, Sarvam AI</i>	<i>Jun 2026</i>
– Turns confusing public-service notices into step-by-step plans in 14 Indian languages , voiced in 11.	
DyslexoFly, Code for Bharat S2 (Team of 2)	Live Code
<i>Next.js, Flask, DistilBART, mT5, edge-tts</i>	<i>Jun 2025 – Jul 2025</i>
– Turns PDFs into dyslexia-friendly reading: OpenDyslexic, Hindi and English summaries, 7 neural voices.	
IIT Mandi LaTeX (Lead)	
<i>Overleaf CE, Node.js, Docker</i>	<i>Sep 2026 – Present</i>
– Institute-hosted Overleaf for ~3,500 users with a fair compile queue and a 50-user waiting room.	
– Per-user quotas, nightly off-host backups and an IIT Mandi template pack.	
TinkerBench & Simulation Lab (Solo)	
<i>JavaScript, Canvas, nginx</i>	<i>Sep 2026 – Oct 2026</i>
– Browser electronics bench that solves circuits by nodal analysis every frame and runs Arduino code.	
– 49 interactive physics and chemistry experiments for school labs.	
Faculty Application Portal	Live
<i>Django, SQL</i>	<i>Dec 2024 – Feb 2025</i>
– Replaced Google Forms with a Django portal on IIT Mandi's recruitment site, with an admin dashboard.	
– Automated CV generation and emails for faculty applicants.	

More projects

- **Face EngagementCV** [Code](#): identifies faces in classroom CCTV and scores engagement, on CPU alone.
- **PSCDL 2026** [Code](#): persistent scene-change detection with classical CV and adaptive thresholds.
- **Vidyt AI** [Code](#): cited industrial-safety answers from OSHA, CSB and Indian standards.
- **FlowAI** [Code](#): Gemini multi-agent planner that splits goals, schedules them and sends reminders.
- **PurrPedia** [Code](#): 247-breed cat encyclopedia on Temporal workflows; all 5 audit findings fixed.
- **CareerAutofill AI** [Code](#): agents and a Chrome extension that fill job forms; LoRA fine-tuning.
- **TheBudgetBuddy** [Code](#): plain-English questions over your transactions, via LangChain and FAISS.
- **SecureInvest-AI** [Code](#): fraud-risk scorer for investment pitches, built solo in 48 hours.
- **Malware scanner** [Code](#): static .exe, .pdf and .docx scanning with YARA rules.
- **TheAttendioBot** [Code](#): Telegram bot that tells students how many classes they can skip.
- **Neufin** [Live](#) [Code](#): product catalogue with search, cart, Zod validation and Jest tests.
- **CyberShield**: mule-account detection on 9,082 bank accounts at a 0.89% base rate.
- **PodSense**: four-agent Kubernetes pod observability design, ABB Accelerator 2026.
- **Solar-panel waste policy**: whether battery-recycling rules carry over to solar waste (team of 5).
- **PRAYAS ML labs**: four teaching notebooks, decision trees to DQN, with 60 guided experiments.
- **Campus self-hosting**: six campus services on one Raspberry Pi 5 behind self-updating redirects.
- **Autonomous ROS robot**: ESP32 on a custom 2-layer PCB, steered by a vector-based ROS planner.
- Also: a research-group website, a GCS cell website, and LaTeX study notes for core courses.

ACHIEVEMENTS

AIR 1, Global Rank 11, SAND 2025 (IEEE ICASSP 2026 challenge): ALS severity from voice.

Rank 10, solo against teams, Vehant Vehicle Count Challenge: classical CV only, MAE 16.4.

Finalist, Pune International Agri Hackathon 2026, Geospatial Crop Survey track.

3rd place, AiHack India 2025, hosted at IIT Mandi by Aiful Corporation, Japan.

Top 1.5% of 262,120 teams, Adobe India Hackathon 2025, as team leader.

Emerald of Gymkhana Award (2025): best GCS Cell volunteer, Dean of Students' Office.

National Semi-Finalist, Tata Imagination Challenge 2024.

1st place, Space Code Challenge 2023, Space Technology & Astronomy Cell, IIT Mandi.

20+ hackathons in 2025–26, incl. Amazon ML Challenge, IDBI Innovate (team lead), ABB Accelerator.

POSITIONS OF RESPONSIBILITY

- UG Academic Affairs Secretary, Student Gymkhana, IIT Mandi** *Jul 2026 – Present*
- Elected student lead for academics for ~**3,500 undergraduates**; head the UG Academic Council.
 - Picked **12 branch councillors** and ran Batch 2026's orientation across **15 branch sessions**.
- Batch 2023 Academic Representative & Academic Guide** *2024 – 2026*
- Linked **500+ students** with faculty; co-ran the first post-COVID offline curriculum session.
 - Hired the council's web team through a live build: 8 candidates, 4 selected.
- Head of Technical Team, Exodia 2026 (Annual Cultural Fest)** *Apr 2026*
- Ran the fest's tech, from ticketing and the gate to the website's performance.
- Instructor, HIMSHIKHAR (CCE IIT Mandi × Masai) & PRAYAS 4.0** *Jun 2026 – Jul 2026*
- Taught **22 classes (40+ hours)** from DBMS to GenAI, to learners from Class 9 to over 80.
 - Ran an 18-hour ML course with 4 Colab labs, leading **8 TAs**.
- UG Head Teaching Assistant, Machine Learning**
- Ran course operations for **600+ students** across a semester.
- Mentoring & Clubs** *2023 – Present*
- GCS and SAE mentor; STAC core member (**100+** hackathon teams); JEE mentor at Physics Wallah.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C++, C, Java, Kotlin, SQL, Rust, R, HTML/CSS, LaTeX
Backend: FastAPI, Django, Flask, Node.js, Express, Fastify, Hono, tRPC, REST APIs, WebSockets
Frontend & Mobile: React, Next.js, Tailwind CSS, Vite, Astro, Tauri, React Native, Android (Kotlin)
ML & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, OpenCV, Hugging Face, LoRA, RAG, LangChain
ML Areas: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Time-Series Forecasting
Data & Cloud: PostgreSQL, Redis, MongoDB, SQLite, Prisma, Docker, AWS, Vercel, Firebase, Linux, CI/CD, Git